

Linear inequalities



1 State whether the following inequalities are represented using a solid line or a dashed line on a graph:

a $y - 3x > 5$

b $x \geq 4$

c $-2y + x \leq -10$

d $y + 3x > 7$

e $-y - 6x > 0$

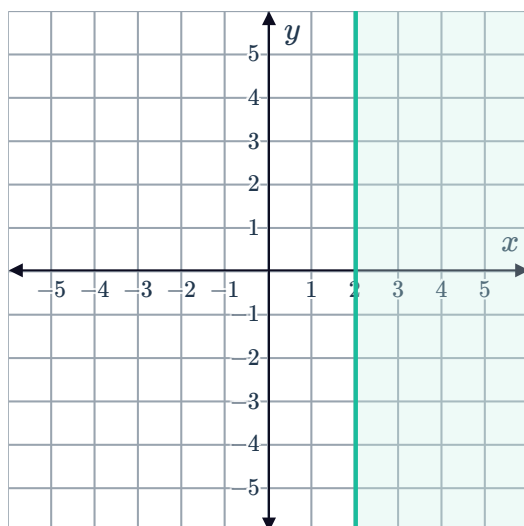
f $y + 3x \leq -10$

g $y \geq -2$

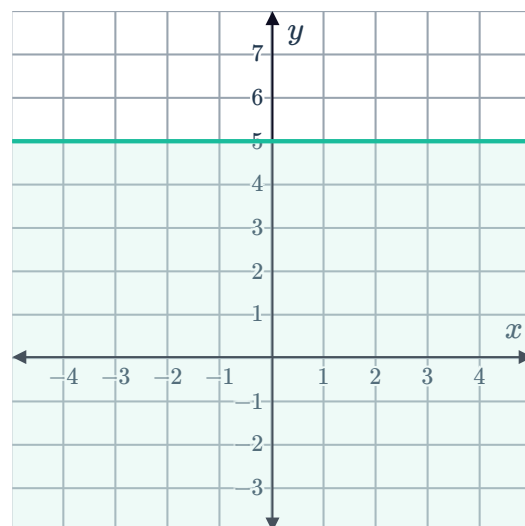
h $3y + 3x > 4$

2 Write the inequality that describes the following shaded regions:

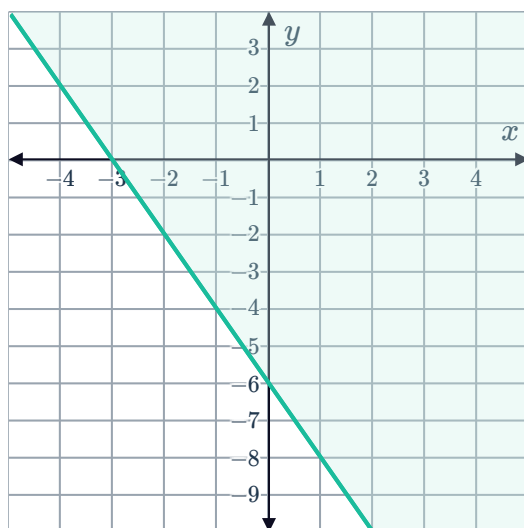
a



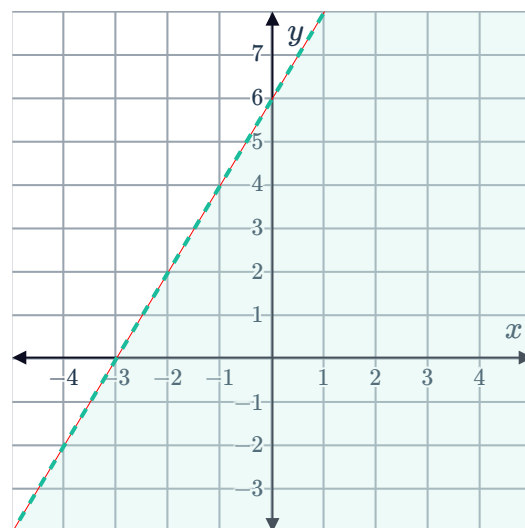
b



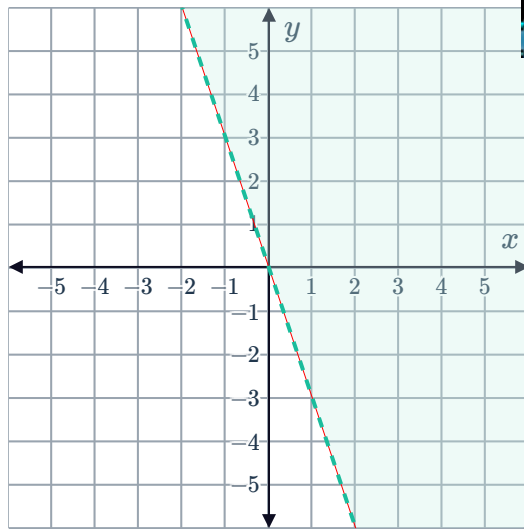
c



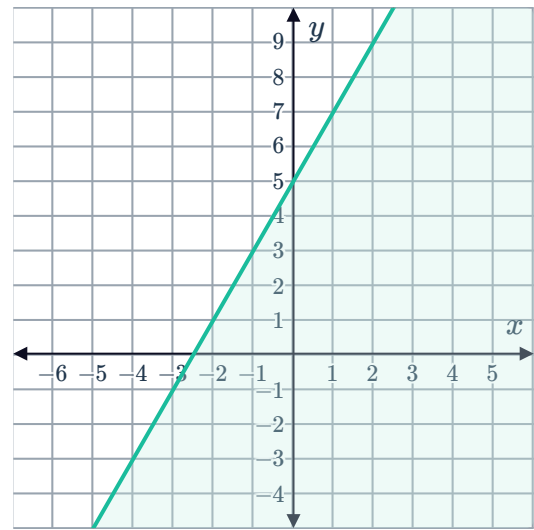
d



e



f



- 3** Is $(6, 20)$ a solution of $y \geq 4x + 5$?
- 4** Is $(3, 2)$ a solution of $3x + 2y \geq 12$?
- 5** Determine which point satisfies each of the following inequalities:
- a** $x - y > 7$
A $(-1, -7)$ **B** $(8, -5)$ **C** $(13, 6)$ **D** $(0, 0)$
- b** $y > 2x$
A $(9, 15)$ **B** $(-1, -2)$ **C** $(1, 4)$ **D** $(0, 0)$
- c** $y > -10x - 2$
A $(5, -52)$ **B** $(-1, 3)$ **C** $(-3, 26)$ **D** $(1, -11)$

Graphing linear inequalities

- 6** Describe the three steps required to graph the inequality $y + 7x \geq 5$.

7 Consider the inequality $y < -3x - 6$.



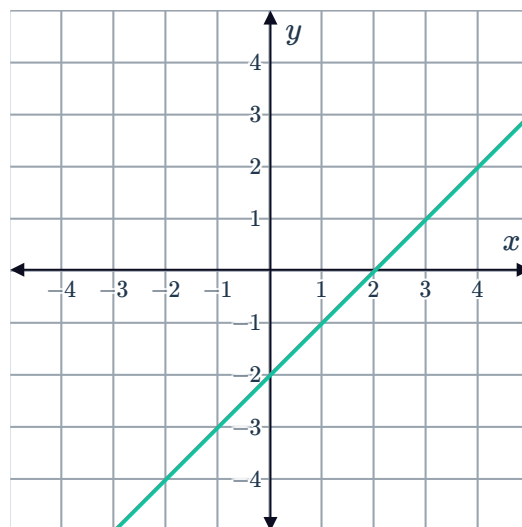
- a Solve for the x and y -intercepts of the line $y = -3x - 6$.
- b Determine which of the following points satisfy the inequality $y < -3x - 6$:
A $(-7, 2)$ **B** $(-6, 3)$ **C** $(3, -2)$ **D** $(-6, -3)$
- c Hence sketch the graph of $y < -3x - 6$.
- d Do the points on the line satisfy the inequality?

8 Consider the inequality $y \leq -2x + 2$.

- a Solve for the x and y intercepts of the line $y = -2x + 2$.
- b Determine which of the following points satisfies the inequality $y \leq -2x + 2$:
A $(2, 3)$ **B** $(3, -6)$ **C** $(4, -2)$ **D** $(1, 2)$
- c Hence sketch the graph of $y \leq -2x + 2$.
- d Do the points on the line satisfy the inequality?

9 The solutions to the equation $y = x - 2$ are plotted on the following graph:

- a Graph the solutions to $y > x - 2$.
- b Graph the solutions to $y \leq x - 2$.



10 Sketch the graphs of the following inequalities:

- | | | | |
|------------------|---------------------|------------------|--------------------|
| a $y \leq x - 2$ | b $5x + 2y \geq 20$ | c $4x + 3y > 12$ | d $6x + 3y > 12$ |
| e $y > 2x - 4$ | f $y \leq 3x - 4$ | g $y < 3x - 9$ | h $y \geq -2x + 4$ |
| i $2x + 3y < 12$ | j $y > 2x - 4$ | k $y \leq x + 5$ | |

11 State the linear inequality that satisfies the following:



- a** A linear inequality has solutions of $(0, -5)$ and $(-1, -9)$ on its boundary line and is not satisfied by $(1, 3)$.
- b** A linear inequality has solutions of $(0, -6)$ and $(-6, -24)$ on its boundary line and is satisfied by $(-3, -10)$.

Applications

12 Luigi has several bottles of lemonade and Pepsi that he wants to take to a picnic, but he only has enough room in his backpack for 9 bottles.

Let x represent the number of bottles of lemonade that he takes to the picnic.

Let y represent the number of bottles of Pepsi that he takes to the picnic.

- a** Write an inequality to represent the situation.
- b** Graph the inequality on a cartesian plane.
- c** Use the graph to determine whether Luigi can take the following combinations of bottles to the picnic:
 - i** 1 bottle of lemonade and 11 bottles of Pepsi
 - ii** 3 bottles of lemonade and 11 bottles of Pepsi
 - iii** 3 bottles of lemonade and 4 bottles of Pepsi
 - iv** 5 bottles of lemonade and 3 bottles of Pepsi

13 Airline passengers are told to be at the gate no less than 15 minutes before a flight's departure. It takes Gwen 30 minutes to get to the airport, and y minutes to go through check in and security, depending on how busy the airport is.

- a** Form an inequality relating x and y , where x represents the number of minutes left until Gwen's flight departs.
- b** Graph the inequality on a cartesian plane.
- c** If the check-in and security process takes 20 minutes, how many minutes before her flight's departure can Gwen leave her house at the latest?

- 14 Roxanne has set aside \$14.00 in her shopping budget for fruit this week. Currently, oranges are on sale for \$1.40 each, while apples are on sale for \$1.75 each.

Let x represent the number of oranges that Roxanne buys.

Let y represent the number of apples that Roxanne buys.

- a Write an inequality to represent the situation.
- b Graph the inequality on a cartesian plane.
- c Use your graph to determine whether Roxanne can buy the following combination of fruits:
 - i 6 oranges and 7 apples
 - ii 3 oranges and 4 apples
 - iii 7 oranges and 4 apples
 - iv 8 oranges and 1 apples